

# HandsMen Threads Salesforce Project Documentation

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**Prepared for Project Review**

Salesforce Platform Developer Project

Data Model Automation Security Apex Testing

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This report presents the HandsMen Threads Salesforce implementation in a concise review format. Each numbered figure panel identifies the exact screenshot that should be captured from the Salesforce developer org before submission.

## 1 Project Overview

HandsMen Threads is a Salesforce solution for a fashion business that manages customers, products, orders, inventory, and marketing campaigns. The project brings operational data into one platform and applies automation to improve customer communication, stock control, loyalty management, and order processing.

The implementation combines custom objects, custom fields, lookup and master detail relationships, formula fields, validation rules, profiles, roles, permission sets, email templates, record triggered flows, scheduled automation, Apex triggers, Batch Apex, and Scheduled Apex.

### Main Business Outcomes

- Send an automatic confirmation message when an order is confirmed.
- Update customer loyalty status from purchase history.
- Notify the inventory team when stock falls below the defined threshold.
- Calculate order totals and deduct stock when confirmed orders are processed.
- Process inventory updates in bulk through scheduled Apex.

### Technology Used

| Area          | Salesforce Feature                                                  |
|---------------|---------------------------------------------------------------------|
| Data          | Custom objects, custom fields, formula fields, relationships        |
| Automation    | Record triggered flows and scheduled flow                           |
| Development   | Apex classes, triggers, handler pattern, Batch Apex, Scheduled Apex |
| Security      | Profiles, roles, users, permission sets, object access              |
| Communication | Classic email templates and email alerts                            |

## 2 Objectives and Implementation Scope

### 2.1 Project Objectives

- Create a structured Salesforce data model for customers, products, orders, inventory, and campaigns.
- Maintain accurate customer, order, and stock information through UI level validation.
- Automate order confirmation, low stock alerts, and loyalty status updates.
- Apply role based access so teams can work only with relevant business records.
- Use Apex for order validation, total calculation, stock deduction, and bulk processing.

### 2.2 Current Implementation Scope

| Area                | Included Implementation                                    | Review Evidence                    |
|---------------------|------------------------------------------------------------|------------------------------------|
| <b>Data model</b>   | Five custom objects with business fields and relationships | Object Manager and Schema Builder  |
| <b>Data quality</b> | Formula fields and validation rules                        | Field and validation rule screens  |
| <b>Security</b>     | Profile, role hierarchy, users, permission set             | Setup security screens             |
| <b>Automation</b>   | Order, stock, and loyalty flows                            | Flow Builder canvases              |
| <b>Development</b>  | Handler, triggers, batch, and scheduler                    | Apex class and trigger screens     |
| <b>Testing</b>      | Sample records, error messages, updates, and emails        | Lightning records and test results |

### 3 Requirement Analysis and Data Model

The data model separates master data from transaction data. Customer and product records store reusable business information. Orders capture transactions, inventory records track stock by product, and marketing campaigns connect promotional activity with customers.

| Custom Object             | Key Fields                                                                 | Business Purpose                                        |
|---------------------------|----------------------------------------------------------------------------|---------------------------------------------------------|
| <b>HandsMen Customer</b>  | Name, First Name, Last Name, Email, Phone, Loyalty Status, Total Purchases | Stores customer identity, contact, and loyalty details. |
| <b>HandsMen Product</b>   | Name, SKU, Price, Stock Quantity                                           | Stores the product catalogue and pricing.               |
| <b>HandsMen Order</b>     | Order Number, Status, Quantity, Total Amount, Customer, Product            | Captures customer purchases and order status.           |
| <b>Inventory</b>          | Inventory Number, Warehouse, Stock Quantity, Product                       | Tracks stock levels for each product.                   |
| <b>Marketing Campaign</b> | Campaign Number, Start Date, End Date, Customer                            | Stores promotional campaigns and customer association.  |

The screenshot shows the Salesforce Developer Edition interface. The top navigation bar includes 'Developer Edition', a search bar, and various utility icons. The main header indicates 'SETUP > OBJECT MANAGER' and 'HandsMen Customer'. A left sidebar lists navigation options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, and Restriction Rules. The 'Fields & Relationships' section shows 11 items, sorted by Field Label. It includes a 'Quick Find' search bar and tabs for 'New', 'Deleted Fields', 'Field Dependencies', and 'Set History Tracking'. The table lists the following fields:

| Field Label            | Field Name         | Data Type          | Controlling Field | Indexed |
|------------------------|--------------------|--------------------|-------------------|---------|
| FirstName              | FirstName__c       | Text(60)           |                   |         |
| FullName               | FullName__c        | Formula (Text)     |                   |         |
| HandsMen Customer Name | Name               | Text(80)           | ✓                 |         |
| Last Modified By       | LastModifiedById   | Lookup(User)       |                   |         |
| Last Name              | LastName__c        | Text(60)           |                   |         |
| Loyalty Status         | Loyalty_Status__c  | Picklist           |                   |         |
| Owner                  | OwnerId            | Lookup(User,Group) | ✓                 |         |
| Phone                  | Phone__c           | Phone              |                   |         |
| Total Purchases        | Total_Purchases__c | Number(18, 0)      |                   |         |

Figure 1 HandsMen Customer object fields including email phone loyalty status and total purchases

The screenshot shows the Salesforce Developer Edition interface for the 'HandsMen Product' object. The top navigation bar is identical to the previous screenshot. The main header indicates 'SETUP > OBJECT MANAGER' and 'HandsMen Product'. The left sidebar is the same. The 'Fields & Relationships' section shows 7 items, sorted by Field Label. It includes a 'Quick Find' search bar and tabs for 'New', 'Deleted Fields', 'Field Dependencies', and 'Set History Tracking'. The table lists the following fields:

| Field Label         | Field Name        | Data Type          | Controlling Field | Indexed |
|---------------------|-------------------|--------------------|-------------------|---------|
| Created By          | CreatedById       | Lookup(User)       |                   |         |
| HandsMen Order Name | Name              | Text(80)           | ✓                 |         |
| Last Modified By    | LastModifiedById  | Lookup(User)       |                   |         |
| Owner               | OwnerId           | Lookup(User,Group) | ✓                 |         |
| Price               | Price__c          | Currency(18, 0)    |                   |         |
| SKU                 | SKU__c            | Text(60)           |                   |         |
| Stock Quantity      | Stock_Quantity__c | Number(18, 0)      |                   |         |

Figure 2 HandsMen Product object fields including SKU price and stock quantity

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER  
**HandsMen Order**

Details

**Fields & Relationships**  
10 Items. Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

|                       |                      |                           |   |
|-----------------------|----------------------|---------------------------|---|
| Created By            | CreatedById          | Lookup(User)              |   |
| Customer Email        | Customer_Email__c    | Email                     |   |
| HandsMen Customer     | HandsMen_Customer__c | Lookup(HandsMen Customer) | ✓ |
| HandsMen Order Number | Name                 | Auto Number               | ✓ |
| HandsMen Product      | HandsMen_Product__c  | Lookup(HandsMen Product)  | ✓ |
| Last Modified By      | LastModifiedById     | Lookup(User)              |   |
| Owner                 | OwnerId              | Lookup(User,Group)        | ✓ |
| Quantity              | Quantity__c          | Number(18, 0)             |   |
| Status                | Status__c            | Picklist                  |   |
| Total Amount          | Total_Amount__c      | Number(18, 0)             |   |

Figure 3 HandsMen Order object fields and relationships including status quantity total amount customer and product

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER  
**Inventory**

Details

**Fields & Relationships**  
7 Items. Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

| FIELD LABEL      | FIELD NAME          | DATA TYPE                       | CONTROLLING FIELD | INDEXED |
|------------------|---------------------|---------------------------------|-------------------|---------|
| Created By       | CreatedById         | Lookup(User)                    |                   |         |
| HandsMen Product | HandsMen_Product__c | Master-Detail(HandsMen Product) |                   | ✓       |
| Inventory Number | Name                | Auto Number                     |                   | ✓       |
| Last Modified By | LastModifiedById    | Lookup(User)                    |                   |         |
| Stock Quantity   | Stock_Quantity__c   | Number(18, 0)                   |                   |         |
| Stock Statues    | Stock_Statues__c    | Formula (Text)                  |                   |         |
| Warehouse        | Warehouse__c        | Text(60)                        |                   |         |

Figure 4 Inventory object fields and relationship including warehouse stock quantity and related product

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER  
**Marketing Campaign**

Details

**Fields & Relationships**  
7 Items. Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

| FIELD LABEL               | FIELD NAME           | DATA TYPE                 | CONTROLLING FIELD | INDEXED |
|---------------------------|----------------------|---------------------------|-------------------|---------|
| Created By                | CreatedById          | Lookup(User)              |                   |         |
| HandsMen Customer         | HandsMen_Customer__c | Lookup(HandsMen Customer) |                   | ✓       |
| Last Date                 | Last_Date__c         | Date                      |                   |         |
| Last Modified By          | LastModifiedById     | Lookup(User)              |                   |         |
| Marketing Campaign Number | Name                 | Auto Number               |                   | ✓       |
| Owner                     | OwnerId              | Lookup(User,Group)        |                   | ✓       |
| Start Date                | Start_Date__c        | Date                      |                   |         |

Figure 6 Marketing Campaign object fields and relationship including campaign dates and related customer

### 3.1 Relationships and Important Fields

| Child Object       | Relationship  | Related Object    | Purpose                                                 |
|--------------------|---------------|-------------------|---------------------------------------------------------|
| Marketing Campaign | Lookup        | HandsMen Customer | Associates a campaign with a customer.                  |
| HandsMen Product   | Lookup        | HandsMen Order    | Connects product information to an order as documented. |
| HandsMen Order     | Lookup        | HandsMen Customer | Links each order to the purchasing customer.            |
| Inventory          | Master Detail | HandsMen Product  | Makes inventory dependent on its product.               |

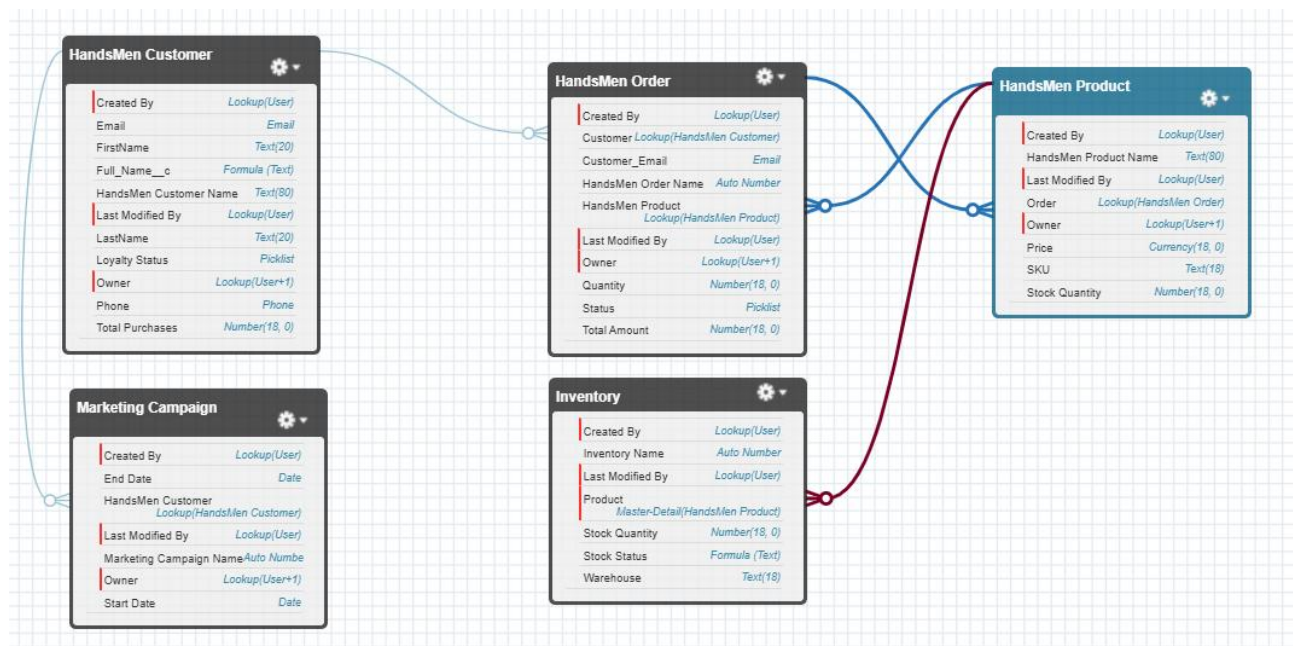


Figure 2 Schema Builder Data Model

## 4 Salesforce Configuration and Data Security

The project uses custom tabs and a dedicated navigation workspace so users can open customer, product, order, inventory, and campaign records from Lightning Experience. Security is controlled through profiles, roles, and a permission set.

| Role                      | Intended Access                                              |
|---------------------------|--------------------------------------------------------------|
| Sales Manager             | Full access to customer and order records.                   |
| Inventory Manager         | Read and edit access to inventory and product records.       |
| Marketing Manager or Team | Read customer records and manage marketing campaign records. |

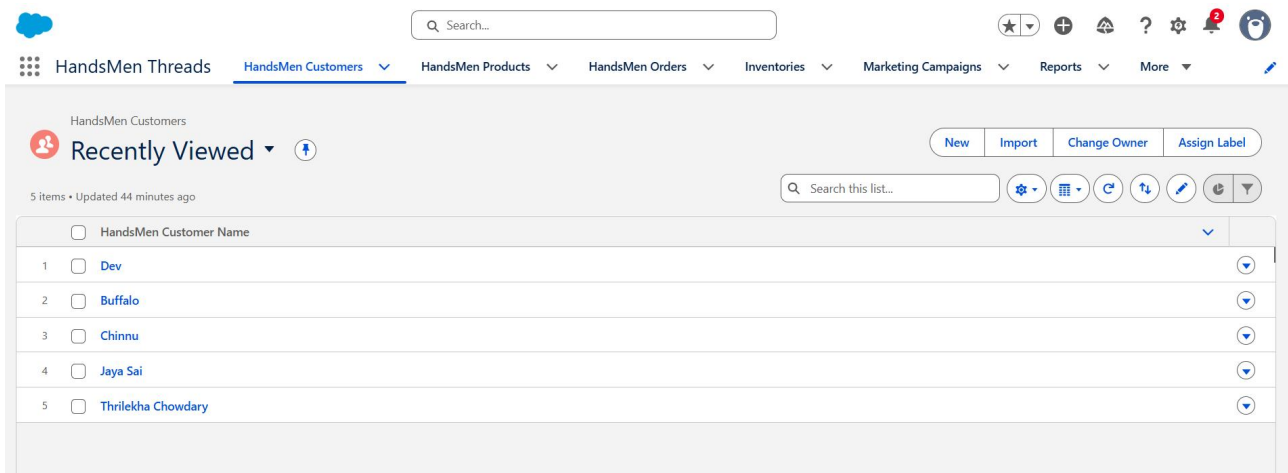


Figure 4 HandsMen Lightning App and Navigation

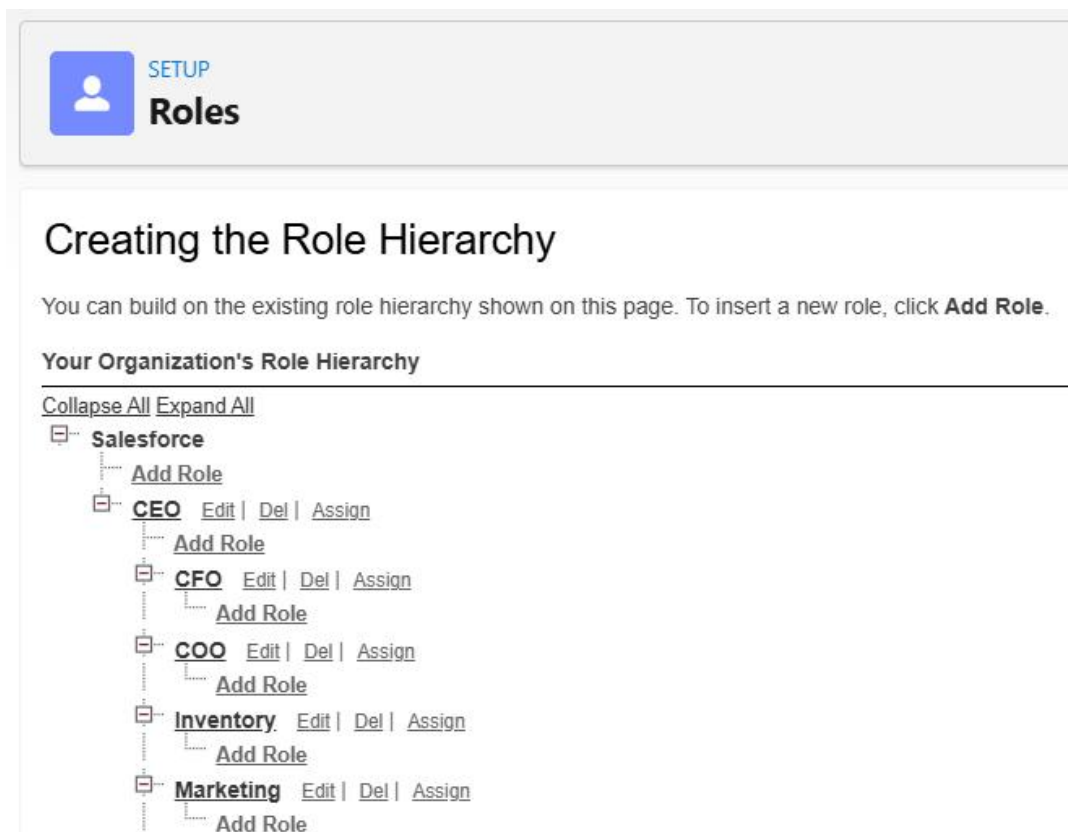


Figure 5 Role Hierarchy

## 4.1 Profiles Users and Permission Sets

| Security Item         | Configuration                             | Purpose                                                                  |
|-----------------------|-------------------------------------------|--------------------------------------------------------------------------|
| <b>Profile</b>        | Marketing cloned from Standard User       | Provides the base Salesforce Platform access.                            |
| <b>User</b>           | Sales user assigned to Sales role         | Supports customer and order activities.                                  |
| <b>User</b>           | Inventory user assigned to Inventory role | Supports product and inventory activities.                               |
| <b>Permission Set</b> | Marketing Permission Sets                 | Adds create, read, edit, and delete access for selected project objects. |

**SETUP**  
**Permission Sets**

Permission Set  
**Permission Platform 1**

Find Settings... Clone Edit Properties Manage Assignments View Summary

Permission Set Overview > Object Settings HandsMen Customers

**HandsMen Customers** Edit

**Tab Settings**

| Available                | Visible                    |
|--------------------------|----------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> i |

**Object Permissions**

| Permission Name    | Enabled                             |
|--------------------|-------------------------------------|
| Read               | <input checked="" type="checkbox"/> |
| Create             | <input checked="" type="checkbox"/> |
| Edit               | <input checked="" type="checkbox"/> |
| Delete             | <input checked="" type="checkbox"/> |
| View All Records   | <input type="checkbox"/>            |
| Modify All Records | <input type="checkbox"/>            |
| View All Fields    | <input type="checkbox"/>            |

Figure 1 HandsMen Customer permissions configured in Permission Platform 1

**SETUP**  
**Permission Sets**

Permission Set  
**Permission Platform 1**

Find Settings... Clone Edit Properties Manage Assignments View Summary

Permission Set Overview > Object Settings HandsMen Orders

**HandsMen Orders** Edit

**Tab Settings**

| Available                | Visible                    |
|--------------------------|----------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> i |

**Object Permissions**

| Permission Name    | Enabled                             |
|--------------------|-------------------------------------|
| Read               | <input checked="" type="checkbox"/> |
| Create             | <input checked="" type="checkbox"/> |
| Edit               | <input checked="" type="checkbox"/> |
| Delete             | <input checked="" type="checkbox"/> |
| View All Records   | <input type="checkbox"/>            |
| Modify All Records | <input type="checkbox"/>            |
| View All Fields    | <input type="checkbox"/>            |

Figure 2 HandsMen Order permissions configured in Permission Platform 1

**SETUP**  
**Permission Sets**

**Permission Platform 1**

Permission Set Overview > Object Settings **HandsMen Products**

**HandsMen Products**

**Tab Settings**

| Available                | Visible                  |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |

**Object Permissions**

| Permission Name    | Enabled                             |
|--------------------|-------------------------------------|
| Read               | <input checked="" type="checkbox"/> |
| Create             | <input checked="" type="checkbox"/> |
| Edit               | <input checked="" type="checkbox"/> |
| Delete             | <input checked="" type="checkbox"/> |
| View All Records   | <input type="checkbox"/>            |
| Modify All Records | <input type="checkbox"/>            |
| View All Fields    | <input type="checkbox"/>            |

Figure 3 HandsMen Products permissions configured in Permission Platform 1

**SETUP**  
**Users**

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other **All**

| <input type="checkbox"/> Action               | Full Name ↑                 | Alias   | Username                                                      | Role      | Active                              | Profile                          |
|-----------------------------------------------|-----------------------------|---------|---------------------------------------------------------------|-----------|-------------------------------------|----------------------------------|
| <input type="checkbox"/> <a href="#">Edit</a> | Chatter Expert              | Chatter | chatty.00daj000017hzvneas.tq2cce5cq7kc@chatter.salesforce.com |           | <input checked="" type="checkbox"/> | Chatter Free User                |
| <input type="checkbox"/> <a href="#">Edit</a> | EPIC_OrgFarm                | OEPIE   | epic.52aadd003d3f@orgfarm.salesforce.com                      |           | <input checked="" type="checkbox"/> | System Administrator             |
| <input type="checkbox"/> <a href="#">Edit</a> | Garimella, Jaya Sai Krishna | jsk     | jaisaikrishna200011@gmail.com                                 | Marketing | <input checked="" type="checkbox"/> | Platform 1                       |
| <input type="checkbox"/> <a href="#">Edit</a> | Kondepati, Thrilekha        | 238     | 238w1a0526.32e2450daa29@agentforce.com                        |           | <input checked="" type="checkbox"/> | System Administrator             |
| <input type="checkbox"/> <a href="#">Edit</a> | Mikaelson, Kol              | kmika   | jaisaikrishna20001114@gmail.com                               | Inventory | <input type="checkbox"/>            | Platform 1                       |
| <input type="checkbox"/> <a href="#">Edit</a> | Mikaelson, Niklaus          | nmika   | thrilekhakondepati1114@gmail.com                              | Sales     | <input checked="" type="checkbox"/> | Platform 1                       |
| <input type="checkbox"/> <a href="#">Edit</a> | User, Integration           | integ   | integration@00daj000017hzvneas.com                            |           | <input checked="" type="checkbox"/> | Analytics Cloud Integration User |
| <input type="checkbox"/> <a href="#">Edit</a> | User, Security              | sec     | insightssecurity@00daj000017hzvneas.com                       |           | <input checked="" type="checkbox"/> | Analytics Cloud Security User    |

Figure 4 Project Users and Assigned Roles

## 5 Data Quality and Business Rules

### 5.1 Formula Fields

| Object                   | Formula Field | Return Type | Formula                                              |
|--------------------------|---------------|-------------|------------------------------------------------------|
| <b>HandsMen Customer</b> | Full Name     | Text        | FirstName__c + " " + LastName__c                     |
| <b>Inventory</b>         | Stock Status  | Text        | IF(Stock_Quantity__c > 10, "Available", "Low Stock") |

### 5.2 Validation Rules

| Object                   | Rule           | Condition                             | Expected Message                                      |
|--------------------------|----------------|---------------------------------------|-------------------------------------------------------|
| <b>HandsMen Order</b>    | Total Amount   | Total_Amount__c <= 0                  | Please enter a correct amount.                        |
| <b>Inventory</b>         | Stock Quantity | Stock_Quantity__c <= 0                | Inventory count cannot be less than or equal to zero. |
| <b>HandsMen Customer</b> | Email          | NOT(CONTAINS(Email__c, "@gmail.com")) | Please enter a correct Gmail address.                 |

SETUP > OBJECT MANAGER

### HandsMen Order

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits

#### Error Condition Formula

**Example:** `Discount_Percent__c > 0.30` [More Examples...](#)  
Display an error if Discount is more than 30%  
If this formula expression is **true**, display the text defined in the Error Message area

Insert Field
Use Formula Assistant
Insert Operator

Total\_Amount\_\_c <= 0

Check Syntax

#### Functions

-- All Function Categories --

ABS
ACOS
ADDMONTHS
AND
ASCII
ASIN

Insert Selected Function

ABS(number)  
Returns the absolute value of a number, a number without its sign

[Help on this function](#)

Figure 1 Total Amount field configuration and successful syntax validation

SETUP > OBJECT MANAGER

### HandsMen Customer

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits

#### Error Condition Formula

**Example:** `Discount_Percent__c > 0.30` [More Examples...](#)  
Display an error if Discount is more than 30%  
If this formula expression is **true**, display the text defined in the Error Message area

Insert Field
Use Formula Assistant
Insert Operator

NOT CONTAINS ( Email\_\_c , "@gmail.com")

Check Syntax

#### Functions

-- All Function Categories --

ABS
ACOS
ADDMONTHS
AND
ASCII
ASIN

Insert Selected Function

ABS(number)  
Returns the absolute value of a number, a number without its sign

[Help on this function](#)

Figure 2 Email formula field configuration and successful syntax validation



Figure 3 Stock Quantity configuration and successful syntax validation

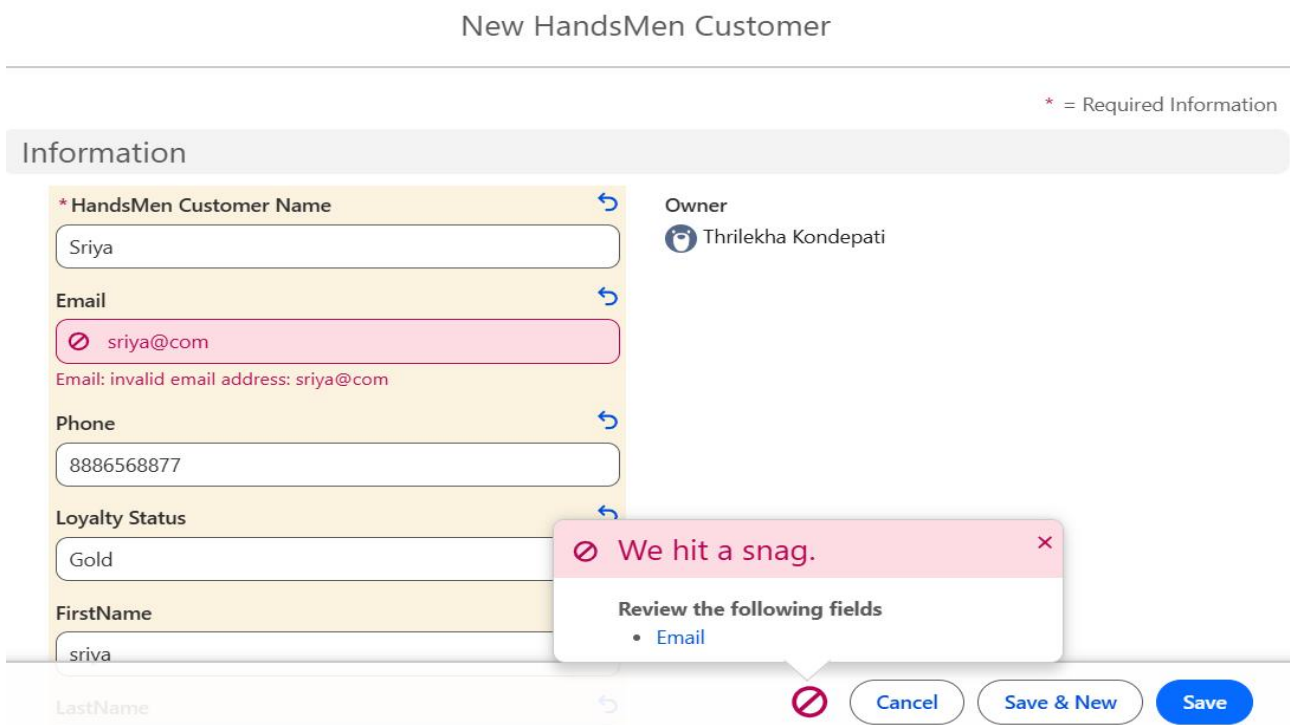
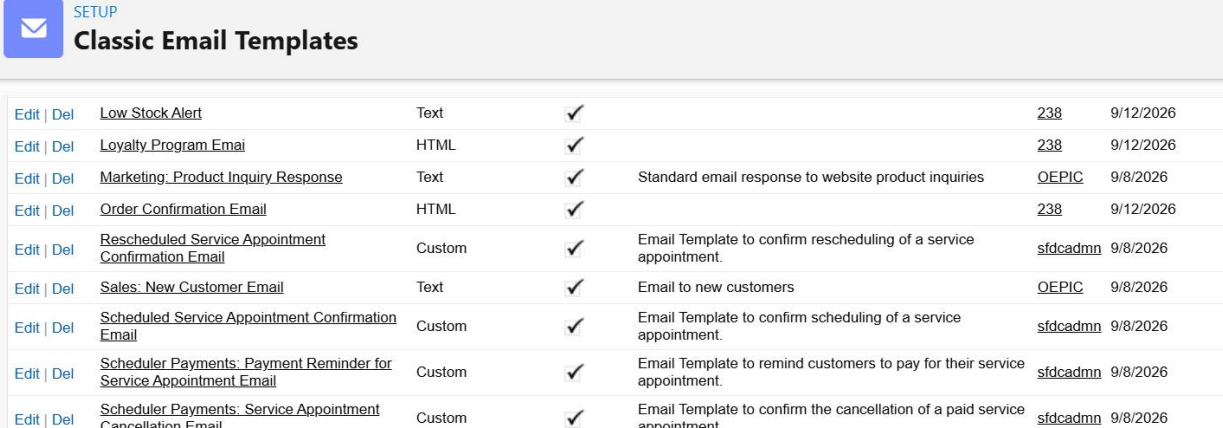


Figure 3 Customer email validation rule preventing an invalid email from being saved

## 6 Email Templates and Process Automation

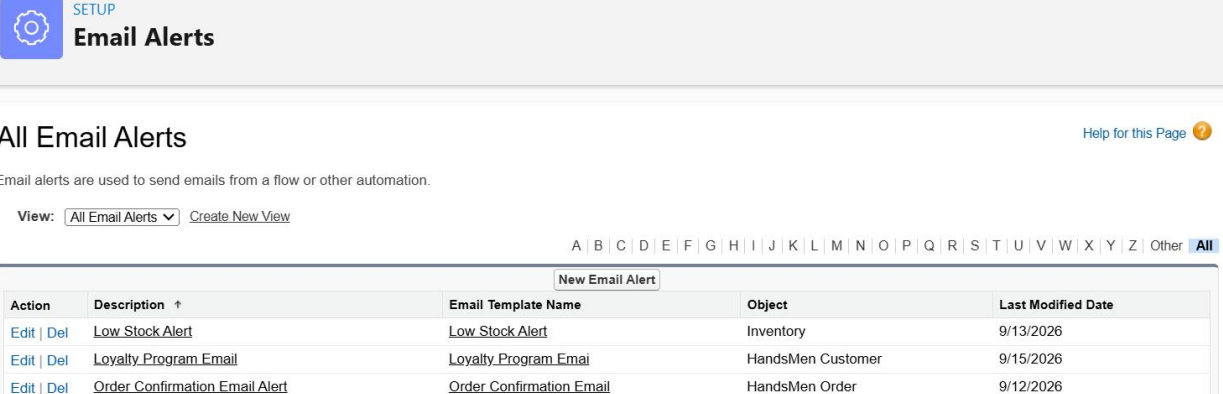
### 6.1 Email Templates and Alerts

| Template              | Type | Used For                                             |
|-----------------------|------|------------------------------------------------------|
| Order Confirmation    | HTML | Sent when an order changes to Confirmed.             |
| Low Stock Alert       | Text | Sent to the inventory team when stock is below five. |
| Loyalty Program Email | HTML | Supports customer loyalty communication.             |



| SETUP Classic Email Templates              |                                                                    |        |   |                                                                           |           |           |
|--------------------------------------------|--------------------------------------------------------------------|--------|---|---------------------------------------------------------------------------|-----------|-----------|
| <a href="#">Edit</a>   <a href="#">Del</a> | Low Stock Alert                                                    | Text   | ✓ |                                                                           | 238       | 9/12/2026 |
| <a href="#">Edit</a>   <a href="#">Del</a> | Loyalty Program Email                                              | HTML   | ✓ |                                                                           | 238       | 9/12/2026 |
| <a href="#">Edit</a>   <a href="#">Del</a> | Marketing: Product Inquiry Response                                | Text   | ✓ | Standard email response to website product inquiries                      | OEPIE     | 9/8/2026  |
| <a href="#">Edit</a>   <a href="#">Del</a> | Order Confirmation Email                                           | HTML   | ✓ |                                                                           | 238       | 9/12/2026 |
| <a href="#">Edit</a>   <a href="#">Del</a> | Rescheduled Service Appointment Confirmation Email                 | Custom | ✓ | Email Template to confirm rescheduling of a service appointment.          | sfdcadmin | 9/8/2026  |
| <a href="#">Edit</a>   <a href="#">Del</a> | Sales: New Customer Email                                          | Text   | ✓ | Email to new customers                                                    | OEPIE     | 9/8/2026  |
| <a href="#">Edit</a>   <a href="#">Del</a> | Scheduled Service Appointment Confirmation Email                   | Custom | ✓ | Email Template to confirm scheduling of a service appointment.            | sfdcadmin | 9/8/2026  |
| <a href="#">Edit</a>   <a href="#">Del</a> | Scheduler Payments: Payment Reminder for Service Appointment Email | Custom | ✓ | Email Template to remind customers to pay for their service appointment.  | sfdcadmin | 9/8/2026  |
| <a href="#">Edit</a>   <a href="#">Del</a> | Scheduler Payments: Service Appointment Cancellation Email         | Custom | ✓ | Email Template to confirm the cancellation of a paid service appointment. | sfdcadmin | 9/8/2026  |

Figure 10 Email Templates List



| SETUP Email Alerts                                                            |                                |                                     |                        |                                    |
|-------------------------------------------------------------------------------|--------------------------------|-------------------------------------|------------------------|------------------------------------|
| All Email Alerts <a href="#">Help for this Page</a>                           |                                |                                     |                        |                                    |
| Email alerts are used to send emails from a flow or other automation.         |                                |                                     |                        |                                    |
| View: <a href="#">All Email Alerts</a> <a href="#">Create New View</a>        |                                |                                     |                        |                                    |
| A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other <a href="#">All</a> |                                |                                     |                        |                                    |
| New Email Alert                                                               |                                |                                     |                        |                                    |
| <a href="#">Action</a>                                                        | <a href="#">Description</a> ↑  | <a href="#">Email Template Name</a> | <a href="#">Object</a> | <a href="#">Last Modified Date</a> |
| <a href="#">Edit</a>   <a href="#">Del</a>                                    | Low Stock Alert                | Low Stock Alert                     | Inventory              | 9/13/2026                          |
| <a href="#">Edit</a>   <a href="#">Del</a>                                    | Loyalty Program Email          | Loyalty Program Email               | HandsMen Customer      | 9/15/2026                          |
| <a href="#">Edit</a>   <a href="#">Del</a>                                    | Order Confirmation Email Alert | Order Confirmation Email            | HandsMen Order         | 9/12/2026                          |

Figure 11 Order Confirmation Email Alert

### 6.2 Salesforce Flows

| Flow                    | Type               | Start Condition                     | Action                                    |
|-------------------------|--------------------|-------------------------------------|-------------------------------------------|
| Order Confirmation Flow | Record Triggered   | Order status changes to Confirmed   | Sends the order confirmation email alert. |
| Stock Alert Flow        | Record Triggered   | Inventory stock quantity is below 5 | Sends a low stock email alert.            |
| Loyalty Status Update   | Schedule Triggered | Runs daily                          | Sets Gold, Silver, or Bronze              |

| Flow | Type | Start Condition | Action                |
|------|------|-----------------|-----------------------|
| Flow |      |                 | from Total Purchases. |

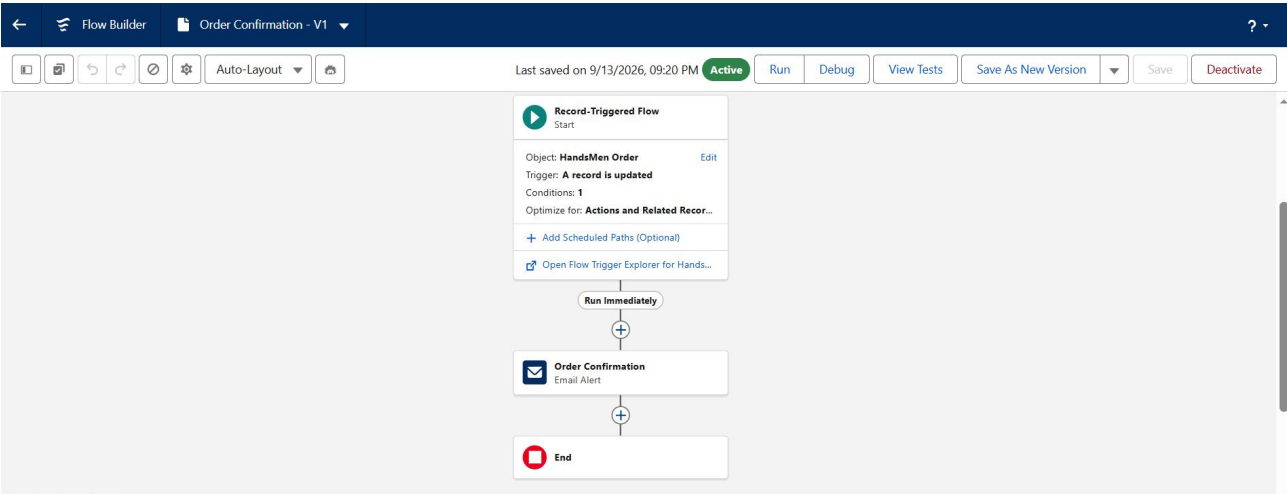


Figure 12 Order Confirmation Flow

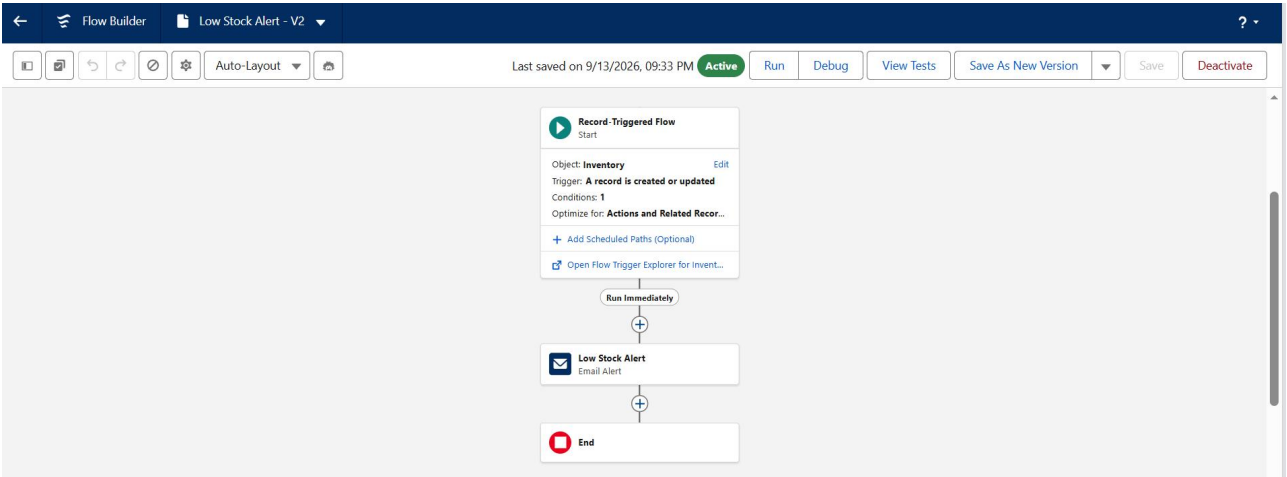


Figure 13 Stock Alert Flow

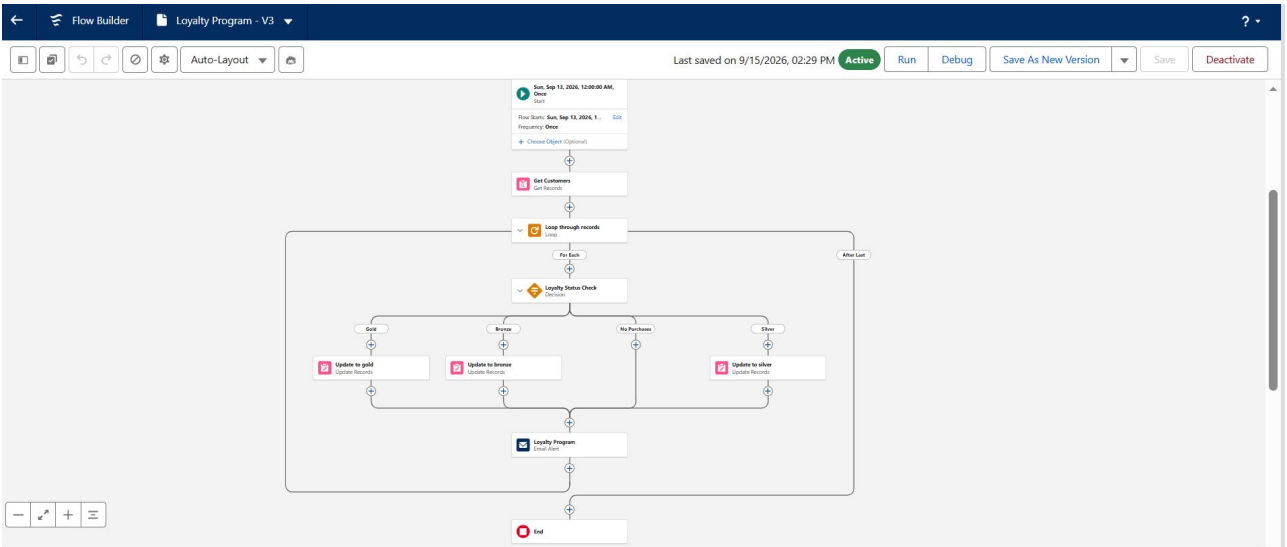


Figure 14 Loyalty Status Update Flow

## 7 Apex Triggers and Handler Logic

Apex is used where record processing requires reusable logic and transaction level enforcement. The order trigger delegates status based quantity validation to OrderTriggerHandler so the trigger remains small and the rule can be maintained in one class.

| Status    | Quantity Rule                     | Outcome                                                        |
|-----------|-----------------------------------|----------------------------------------------------------------|
| Confirmed | Quantity must be greater than 500 | The record is blocked when the quantity is missing or too low. |
| Pending   | Quantity must be greater than 200 | The record is blocked when the quantity is missing or too low. |
| Rejection | Quantity must equal 0             | Rejected orders cannot contain a positive quantity.            |

```
trigger OrderTrigger on HandsMen_Order__c (before insert, before update) {  
    OrderTriggerHandler.validateOrderQuantity(Trigger.new);  
}
```

```
public class OrderTriggerHandler {  
    public static void validateOrderQuantity(List<HandsMen_Order__c> orderList) {  
        for (HandsMen_Order__c order : orderList) {  
            if (order.Status__c == 'Confirmed') {  
                if (order.Quantity__c == null || order.Quantity__c <= 500) {  
                    order.Quantity__c.addError('For Status "Confirmed", Quantity must be more than 500.');                }  
            } else if (order.Status__c == 'Pending') {  
                if (order.Quantity__c == null || order.Quantity__c <= 200) {  
                    order.Quantity__c.addError('For Status "Pending", Quantity must be more than 200.');                }  
            } else if (order.Status__c == 'Rejection') {  
                if (order.Quantity__c == null || order.Quantity__c != 0) {  
                    order.Quantity__c.addError('For Status "Rejection", Quantity must be 0.');                }  
            }  
        }  
    }  
}
```

Figure 15 OrderTriggerHandler Apex Class

```
trigger OrderTrigger on HandsMen_Order__c (before insert, before update) {  
    if (Trigger.isBefore && (Trigger.isInsert || Trigger.isUpdate)) {  
        OrderTriggerHandler.validateOrderQuantity(Trigger.new);  
    }  
}
```

Figure 16 OrderTrigger Apex Trigger

## 7.1 Order Total and Stock Deduction Triggers

| Apex Artifact                  | Event                           | Responsibility                                                                                            |
|--------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Order Total Trigger</b>     | Before insert and before update | Reads the related product price and sets Total Amount to Quantity multiplied by Price.                    |
| <b>Stock Deduction Trigger</b> | After insert and after update   | When an order becomes Confirmed, totals ordered quantity by product and reduces Inventory Stock Quantity. |

The stock deduction design collects product identifiers in a set, aggregates quantities in a map, queries matching inventory records once, and performs one bulk update. This avoids database operations inside the record loop.

```

1  trigger OrderTotalTrigger on HandsMen_Order__c (before insert, before update) {
2      Set<Id> productIds = new Set<Id>();
3
4      for (HandsMen_Order__c order : Trigger.new) {
5          if (order.HandsMen_Product__c != null) {
6              productIds.add(order.HandsMen_Product__c);
7          }
8      }
9
10     Map<Id, HandsMen_Product__c> productMap = new Map<Id, HandsMen_Product__c>(
11         [SELECT Id, Price__c FROM HandsMen_Product__c WHERE Id IN :productIds]
12     );
13
14     for (HandsMen_Order__c order : Trigger.new) {
15         if (order.HandsMen_Product__c != null && productMap.containsKey(order.HandsMen_Product__c)) {
16             HandsMen_Product__c product = productMap.get(order.HandsMen_Product__c);
17             if (order.Quantity__c != null) {
18                 order.Total_Amount__c = order.Quantity__c * product.Price__c;
19             }
20         }
21     }
22 }

```

**Figure 17 Order Total Trigger**

```

trigger StockDeductionTrigger on HandsMen_Order__c (after insert, after update) {

    Set<Id> productIds = new Set<Id>();
    Map<Id, Decimal> quantityByProduct = new Map<Id, Decimal>();

    // Find products whose stock needs to be deducted
    for (HandsMen_Order__c order : Trigger.new) {

        Boolean shouldDeduct = false;

        // New order is Confirmed
        if (Trigger.isInsert &&
            order.Status__c == 'Confirmed') {

            shouldDeduct = true;
        }

        // Existing order changed to Confirmed
        if (Trigger.isUpdate) {

            HandsMen_Order__c oldOrder =
                Trigger.oldMap.get(order.Id);

            if (oldOrder.Status__c != 'Confirmed' &&
                order.Status__c == 'Confirmed') {

                shouldDeduct = true;
            }
        }

        // Check product and quantity
        if (shouldDeduct &&
            order.HandsMen_Product__c != null &&
            order.Quantity__c != null &&
            order.Quantity__c > 0) {

            Id productId = order.HandsMen_Product__c;

            productIds.add(productId);

            Decimal currentQuantity = 0;

            if (quantityByProduct.containsKey(productId)
                currentQuantity =
                    quantityByProduct.get(productId);
            }

            quantityByProduct.put(
                productId,
                currentQuantity + order.Quantity__c
            );
        }
    }

    // Nothing to deduct
    if (productIds.isEmpty()) {
        return;
    }

    // Get inventory
    List<Inventory__c> inventories = [
        SELECT Id,
            Stock_Quantity__c,
            HandsMen_Product__c
        FROM Inventory__c
        WHERE HandsMen_Product__c IN :productIds
        FOR UPDATE
    ];

    List<Inventory__c> inventoriesToUpdate =
        new List<Inventory__c>();

    // Check and deduct stock
    for (Inventory__c inventory : inventories) {

        if (!quantityByProduct.containsKey(
            inventory.HandsMen_Product__c
        )) {
            continue;
        }

        Decimal quantityToDeduct =
            quantityByProduct.get(
                inventory.HandsMen_Product__c
            );

        Decimal currentStock =
            inventory.Stock_Quantity__c == null
            ? 0
            : inventory.Stock_Quantity__c;

        // Do not allow insufficient stock
        if (currentStock < quantityToDeduct) {

            HandsMen_Order__c relatedOrder;

            for (HandsMen_Order__c order : Trigger.new) {
                if (order.HandsMen_Product__c ==
                    inventory.HandsMen_Product__c &&
                    order.Status__c == 'Confirmed') {

                    relatedOrder = order;
                    break;
                }
            }

            if (relatedOrder != null) {
                relatedOrder.addError(
                    'Insufficient inventory. Available stock: ' +
                    currentStock +
                    ', requested quantity: ' +
                    quantityToDeduct + '.'
                );
            }

            continue;
        }

        Decimal remainingStock =
            currentStock - quantityToDeduct;

        Inventory__c inventoryUpdate =
            new Inventory__c();

        inventoryUpdate.Id = inventory.Id;
        inventoryUpdate.Stock_Quantity__c = remainingStock;

        inventoriesToUpdate.add(inventoryUpdate);
    }

    // Update inventory
    if (!inventoriesToUpdate.isEmpty()) {
        update inventoriesToUpdate;
    }
}

```

**Figure 18 Stock Deduction Trigger**

## 8 Batch Apex and Scheduled Processing

InventoryBatchJob implements Database Batchable and Schedulable. It identifies low stock products, applies the restocking logic in manageable batches, and can be scheduled for automatic execution.

| Method or Process          | Responsibility                                         | Expected Evidence                       |
|----------------------------|--------------------------------------------------------|-----------------------------------------|
| <b>start</b>               | Queries products that meet the low stock condition.    | SOQL query and selected fields.         |
| <b>execute</b>             | Adds the restock quantity and updates records in bulk. | Loop, update list, and DML statement.   |
| <b>finish</b>              | Records that inventory synchronization completed.      | Completion debug statement.             |
| <b>Schedulable execute</b> | Starts the batch with a scope of 200.                  | Database executeBatch call.             |
| <b>Scheduled job</b>       | Runs Daily Inventory Sync automatically.               | Scheduled Jobs entry and next run time. |

```
global class InventoryBatchJob
    implements Database.Batchable<SObject>, Schedulable {

    global Database.QueryLocator start(Database.BatchableContext BC) {
        return Database.getQueryLocator(
            'SELECT Id, Stock_Quantity__c ' +
            'FROM HandsMen_Product__c ' +
            'WHERE Stock_Quantity__c < 10'
        );
    }

    global void execute(
        Database.BatchableContext BC,
        List<SObject> records
    ) {
        List<HandsMen_Product__c> productsToUpdate =
            new List<HandsMen_Product__c>();

        for (SObject record : records) {
            HandsMen_Product__c product =
                (HandsMen_Product__c) record;

            product.Stock_Quantity__c += 50;
            productsToUpdate.add(product);
        }

        product.Stock_Quantity__c += 50;
        productsToUpdate.add(product);
    }

    global void finish(Database.BatchableContext BC) {
        System.debug('Inventory Sync Completed');
    }

    global void execute(SchedulableContext SC) {
        Database.executeBatch(new InventoryBatchJob(), 200);
    }
}
```

Figure 19 InventoryBatchJob Apex Class


**SETUP**  
**Scheduled Jobs**

### All Scheduled Jobs

The All Scheduled Jobs page lists all of the jobs scheduled by your users. Multiple job types may display on this page. You can delete scheduled jobs if you have the permission to do so.

**Percentage of Scheduled Jobs Used: 1%**  
 You have currently used 1 scheduled Apex jobs out of an allowed organization limit of 100 active or scheduled jobs. To learn about how this limit is calculated and what contributes to it see the [Lightning Platform Apex Limits](#) topic.

View: All Scheduled Jobs [Create New View](#)

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | **All**

| Action                                                                   | Job Name ↑           | Submitted By          | Submitted          | Started             | Next Scheduled Run  | Type           | Cron Trigger ID |
|--------------------------------------------------------------------------|----------------------|-----------------------|--------------------|---------------------|---------------------|----------------|-----------------|
| <a href="#">Manage</a>   <a href="#">Del</a>   <a href="#">Pause Job</a> | Daily Inventory Sync | Kondepati, Thirilekha | 9/19/2026, 7:24 AM | 9/20/2026, 12:03 AM | 9/21/2026, 12:00 AM | Scheduled Apex | 08eaj0000112hca |

Figure 20 Scheduled Inventory Job

## 9 User Interface and Sample Records

Sample records should demonstrate that the configured fields, relationships, and automation are available to business users in Lightning Experience. Use realistic but non-sensitive data and keep the browser zoom between 80 and 100 percent so labels and values remain readable.

| Record                    | Recommended Fields to Show                                         |
|---------------------------|--------------------------------------------------------------------|
| <b>HandsMen Customer</b>  | Full name, email, phone, loyalty status, total purchases           |
| <b>HandsMen Product</b>   | Product name, SKU, price, stock quantity                           |
| <b>HandsMen Order</b>     | Order number, customer, product, status, quantity, total amount    |
| <b>Inventory</b>          | Inventory number, product, warehouse, stock quantity, stock status |
| <b>Marketing Campaign</b> | Campaign number or name, dates, related customer                   |

HandsMen Threads **HandsMen Customers** HandsMen Products HandsMen Orders Inventories Marketing Campaigns

HandsMen Customer  
**Thrilekha Chowdary**

Related **Details**

|                                                                                       |                                                                       |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| HandsMen Customer Name<br>Thrilekha Chowdary                                          | Owner<br><a href="#">Thrilekha Kondepati</a>                          |
| Email<br><a href="mailto:jaisaikrishna2000@gmail.com">jaisaikrishna2000@gmail.com</a> |                                                                       |
| Phone<br>(888) 656-8877                                                               |                                                                       |
| Loyalty Status<br>Bronze                                                              |                                                                       |
| FirstName<br>thrilekha                                                                |                                                                       |
| LastName<br>k                                                                         |                                                                       |
| FullName<br>thrilekha k                                                               |                                                                       |
| Total Purchases<br>200                                                                |                                                                       |
| Created By<br><a href="#">Thrilekha Kondepati</a> , 9/15/2026, 2:17 AM                | Last Modified By<br><a href="#">OrgFarm EPIC</a> , 9/15/2026, 2:32 AM |

HandsMen Threads **HandsMen Products** HandsMen Customers HandsMen Orders Inventories Marketing Campaigns

HandsMen Product  
**T-Shirt Cloth**

Related **Details**

|                                                                        |                                                                              |
|------------------------------------------------------------------------|------------------------------------------------------------------------------|
| HandsMen Order Name<br>T-Shirt Cloth                                   | Owner<br><a href="#">Thrilekha Kondepati</a>                                 |
| SKU                                                                    |                                                                              |
| Price<br>\$10                                                          |                                                                              |
| Stock Quantity                                                         |                                                                              |
| Created By<br><a href="#">Thrilekha Kondepati</a> , 9/12/2026, 9:11 AM | Last Modified By<br><a href="#">Thrilekha Kondepati</a> , 9/20/2026, 2:30 AM |

Figure 21 Customer and Product Sample Records



HandsMen Threads

HandsMen Customers

HandsMen Products

HandsMen Orders

Inventories

Marketing Campaigns



HandsMen Order

O-0013

Related

Details

HandsMen Order Number

O-0013

HandsMen Product

T-Shirt Cloth

HandsMen Customer

Thrilekha Chowdary

Status

Pending

Quantity

90

Total Amount

900

Customer Email

thrilekhakondepati@gmail.com

Created By



Thrilekha Kondepati

9/19/2026, 7:53 AM

Owner



Thrilekha Kondepati

Last Modified By



Thrilekha Kondepati

9/20/2026, 8:55 AM

Figure 22 Order and Inventory Sample Records

## 10 Functional Testing Evidence

Functional testing should show both configuration and business outcomes. Capture the action and its result, and use before and after screenshots where a value changes.

| Test Case                        | Action                                                                 | Expected Result                                        | Evidence               |
|----------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|------------------------|
| <b>Order quantity validation</b> | Save a status and quantity combination that violates the handler rule. | Save is blocked with the Apex error.                   | Error on Order page    |
| <b>Order total</b>               | Create or edit an order with product and quantity.                     | Total Amount equals Quantity multiplied by Price.      | Order values           |
| <b>Stock deduction</b>           | Change an order from Pending to Confirmed.                             | Inventory decreases once by the ordered quantity.      | Before and after stock |
| <b>Order confirmation</b>        | Confirm an order.                                                      | Customer receives the confirmation email.              | Email inbox            |
| <b>Low stock alert</b>           | Set stock below 5.                                                     | Inventory manager receives an alert.                   | Email inbox            |
| <b>Loyalty update</b>            | Run or debug the scheduled flow with purchase values.                  | Customer loyalty changes to Gold, Silver, or Bronze.   | Flow debug and record  |
| <b>Batch processing</b>          | Execute InventoryBatchJob.                                             | Low stock records are processed and the job completes. | Apex Jobs and record   |

HandsMen Order Number

O-0009

Owner

 Thrilekha Kondepoti

HandsMen Product

 T-Shirt Cloth 

HandsMen Customer

 Chinnu 

Status

Confirmed 

Quantity

300

Total Amount

1,500

 We hit a snag. 

Review the errors on this page.

- Insufficient inventory. Available stock: 10, requested quantity: 300.

\* Customer Email

thrilekhakondepoti@gmail.com

Figure 23 Order Validation Error

## 10.1 Automation Test Results

HandsMen Product

 **T-Shirt Cloth**

Related

Details

HandsMen Order Name

T-Shirt Cloth

SKU

Price

\$10

Stock Quantity

505

Created By

 [Thrilekha Kondepati](#), 9/20/2026, 3:26 AM

Owner

 [Thrilekha Kondepati](#)

Last Modified By

 [Thrilekha Kondepati](#), 9/20/2026, 3:26 AM

HandsMen Order

 **O-0005**

Related

Details

HandsMen Order Number

O-0005

HandsMen Product

[T-Shirt Cloth](#)

HandsMen Customer

[Buffalo](#)

Status

Confirmed

Quantity

200

Total Amount

1,000

Customer Email

[thrilekhakondepati@gmail.com](#)

Created By

 [Thrilekha Kondepati](#), 9/15/2026, 3:42 AM

Owner

 [Thrilekha Kondepati](#)

Last Modified By

 [Thrilekha Kondepati](#), 9/19/2026, 8:12 AM

Figure 24 Order Total Calculation

Inventory

 **I -0006**

Related

Details

Inventory Number

I -0006

HandsMen Product

[T-Shirt Cloth](#)

Stock Quantity

10

Stock Statues

Low Stock

Warehouse

Created By

 [Thrilekha Kondepati](#), 9/19/2026, 8:17 AM

Last Modified By

 [Thrilekha Kondepati](#), 9/19/2026, 8:18 AM

|                                                                                                                                                                                                                                                                                                       |                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <div> <div> <div></div> <div>HandsMen Threads</div> </div> <div> <div></div> <div>HandsMen Customers</div> </div> <div> <div></div> <div>HandsMen Products</div> </div> <div> <div></div> <div>HandsMen Order "O-0025" was created.</div> </div> <div> <div></div> <div>Campaigns</div> </div> </div> |                                                                            |
| <div> <div> <div></div> <div>HandsMen Order</div> </div> <div>O-0025</div> </div>                                                                                                                                                                                                                     |                                                                            |
| Related                                                                                                                                                                                                                                                                                               | Details                                                                    |
| HandsMen Order Number                                                                                                                                                                                                                                                                                 | Owner                                                                      |
| O-0025                                                                                                                                                                                                                                                                                                | <div> <div></div> <div>Thrilekha Kondepati</div> </div>                    |
| HandsMen Product                                                                                                                                                                                                                                                                                      |                                                                            |
| T-Shirt Cloth                                                                                                                                                                                                                                                                                         |                                                                            |
| HandsMen Customer                                                                                                                                                                                                                                                                                     |                                                                            |
| Jaya                                                                                                                                                                                                                                                                                                  |                                                                            |
| Status                                                                                                                                                                                                                                                                                                |                                                                            |
| Pending                                                                                                                                                                                                                                                                                               |                                                                            |
| Quantity                                                                                                                                                                                                                                                                                              |                                                                            |
| 300                                                                                                                                                                                                                                                                                                   |                                                                            |
| Total Amount                                                                                                                                                                                                                                                                                          |                                                                            |
| 3,000                                                                                                                                                                                                                                                                                                 |                                                                            |
| Customer Email                                                                                                                                                                                                                                                                                        |                                                                            |
| thrilekhakondepati@gmail.com                                                                                                                                                                                                                                                                          |                                                                            |
| Created By                                                                                                                                                                                                                                                                                            | Last Modified By                                                           |
| <div> <div></div> <div>Thrilekha Kondepati</div> </div> 10/3/2026, 5:49 AM                                                                                                                                                                                                                            | <div> <div></div> <div>Thrilekha Kondepati</div> </div> 10/3/2026, 5:49 AM |

 **Thrilekha Kondepati** <email@00daj000017hzhvneas.sfcustomeremail.com>  
to me ▾

---

*Dear Prem,*  
*Your order #O-0022 has been confirmed!*  
*Thank you for shopping with us.*  
*Best Regards,*  
*Sales Team*

## Low Stock Alert Email

External

Inbox x



**Thrilekha Kondepati** <email@00daj000017hzvneas.sfcustomeremail.com>

to me ▼

Dear Inventory Manager,

This is to inform you that the stock for the following product is running low:

Product Name: Pant Cloth

Current Stock Quantity: 4

Please take the necessary steps to restock this item immediately.

Best Regards,

Inventory Monitoring System

Figure 26 Automated Email Evidence

## 10.2 Scheduled Automation and Batch Results

HandsMen Customer

 **Thrilekha Chowdary**

Related

Details

HandsMen Customer Name

Thrilekha Chowdary

Email

[jaisaikrishna2000@gmail.com](mailto:jaisaikrishna2000@gmail.com)

Phone

[\(888\) 656-8877](tel:(888)656-8877)

Loyalty Status

Bronze

FirstName

thrilekha

LastName

k

FullName

thrilekha k

Total Purchases

200

Created By

 [Thrilekha Kondepati](#), 9/15/2026, 2:17 AM

Owner

 [Thrilekha Kondepati](#)

Last Modified By

 [OrgFarm EPIC](#), 9/15/2026, 2:32 AM

Figure 27 Loyalty Status Update Result

|  <b>SETUP</b><br><b>Apex Jobs</b> |                     |            |           |               |               |                   |          |                      |                     |                   |             |             |
|--------------------------------------------------------------------------------------------------------------------|---------------------|------------|-----------|---------------|---------------|-------------------|----------|----------------------|---------------------|-------------------|-------------|-------------|
| View: <span>All</span> <a href="#">Create New View</a>                                                             |                     |            |           |               |               |                   |          |                      |                     |                   |             |             |
| Action                                                                                                             | Submitted Date      | Job Type   | Status    | Status Detail | Total Batches | Batches Processed | Failures | Submitted By         | Completion Date     | Apex Class        | Apex Method | Apex Job ID |
|                                                                                                                    | 10/3/2026, 12:07 AM | Batch Apex | Completed |               | 0             | 0                 | 0        | Kondepati, Thrilekha | 10/3/2026, 12:07 AM | InventoryBatchJob |             | 707aj0000   |
|                                                                                                                    | 10/2/2026, 12:02 AM | Batch Apex | Completed |               | 0             | 0                 | 0        | Kondepati, Thrilekha | 10/2/2026, 12:02 AM | InventoryBatchJob |             | 707aj0000   |
|                                                                                                                    | 10/1/2026, 12:02 AM | Batch Apex | Completed |               | 0             | 0                 | 0        | Kondepati, Thrilekha | 10/1/2026, 12:02 AM | InventoryBatchJob |             | 707aj0000   |
|                                                                                                                    | 9/30/2026, 12:03 AM | Batch Apex | Completed |               | 0             | 0                 | 0        | Kondepati, Thrilekha | 9/30/2026, 12:03 AM | InventoryBatchJob |             | 707aj0000   |
|                                                                                                                    | 9/29/2026, 12:07 AM | Batch Apex | Completed |               | 0             | 0                 | 0        | Kondepati, Thrilekha | 9/29/2026, 12:07 AM | InventoryBatchJob |             | 707aj0000   |
|                                                                                                                    | 9/28/2026, 12:08 AM | Batch Apex | Completed |               | 0             | 0                 | 0        | Kondepati, Thrilekha | 9/28/2026, 12:08 AM | InventoryBatchJob |             | 707aj0000   |
|                                                                                                                    | 9/27/2026, 12:08 AM | Batch Apex | Completed |               | 0             | 0                 | 0        | Kondepati, Thrilekha | 9/27/2026, 12:08 AM | InventoryBatchJob |             | 707aj0000   |

Figure 28 Batch Job Execution Result

## 11 Conclusion

The HandsMen Threads Salesforce project provides a practical CRM and operations solution for a fashion business. It organizes customer, product, order, inventory, and campaign data; applies validation and formula logic; controls access with Salesforce security features; automates customer and stock communication; and uses Apex for business rules and bulk processing.

Together, the configuration, automation, development, and testing evidence demonstrate core Salesforce Platform Developer skills. The completed screenshot set will give the reviewer a clear path from the data model and security design to working automation and functional outcomes.